Start with the exposure you want to change

LECTURE 12 - REVIEW NOTES

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Before adding a swap, I want you to identify the cash flow you already have. A borrower pays interest; a bond investor receives it. The same change in market rates can therefore help one and hurt the other. In this lecture, we move from managing a company's debt payments to managing the income of a Mexican bond portfolio.

Keep the original contract; add the swap

Firm A owes SOFR + 1% and wants a fixed borrowing cost. It pays 6% fixed and receives SOFR through a swap. Firm B owes 8% fixed, pays SOFR through the same direct swap and receives 6%. Assume matched notionals, benchmarks, maturities and semiannual dates.

Firm	Debt rate	Swap pays	Swap receives	Final cost
A	SOFR + 1%	6%	SOFR	7% fixed
B	8%	SOFR	6%	SOFR + 2%

Final borrowing rate = debt rate + swap pay rate - swap receive rate

For A, the SOFR terms cancel. For B, the fixed receipt reduces its original fixed debt cost, leaving floating exposure. These results show a change in exposure; they do not by themselves establish savings against another borrowing alternative.

A bank can stand between the two firms

Consider a separate example. A owes 6%, pays SOFR + 0.5% to a bank and receives 4%. B owes SOFR + 1%, pays 4.5% to that bank and receives SOFR. With equal notionals and matched dates, A ends at SOFR + 2.5%, while B ends at 5.5% fixed.

Bank net rate = (SOFR + 0.5%) + 4.5% - 4% - SOFR = 1%

The bank earns a fixed annual spread on one common notional, before expenses and default losses. It does not depend on the SOFR path. Its dollar cash flow still depends on the notional and the length of the payment period.

Keep the signs consistent

Net swap cash flow always means received minus paid. For a borrower, subtract that net receipt from the debt payment. For an investor, add it to the bond receipt. A negative swap cash flow is a payment, not automatically a failed hedge.

The portfolio manager wants predictable income

Our portfolio holds one Mexican floating-rate bond. If TIIE falls, the bond keeps its contractual spread, but its future coupons become smaller. The manager wants to lock the peso income instead of depending on future rate resets.

Bond term	Teaching case
Purchase price / face	MXN 1,000 / MXN 1,000
Annual coupon rate	TIIE + 2%
Payment dates	t=0.5, 1.0, 1.5, 2.0, 2.5 and 3.0
Principal repayment	MXN 1,000 at t=3.0
TIIE observed at t=0	8%

All rates are nominal annual rates with semiannual payments. We use the rate observed at the start of each semester: the t=0 observation determines the payment at t=0.5. This is our simplified classroom reset convention, rather than a description of every actual TIIE-linked contract. Ignore fees, taxes and default, and hold the bond to maturity.

Receive fixed and pay the benchmark

The manager enters an IRS on MXN 1,000, receiving 8% fixed and paying TIIE. The swap has the same three-year maturity and the same reset and payment dates as the bond. The 8% fixed rate is an assumed teaching quote. The IRS exchanges interest only; its notional is not repaid at maturity.

Combined annual income = (TIIE + 2%) + 8% - TIIE = 10%

The floating coupon receipt is offset by the floating swap payment. What remains is the 8% fixed receipt plus the bond's 2% spread. This fixes interest income at MXN 50 every semester. At maturity, add the bond's MXN 1,000 principal.

Write the cash flows before inserting a scenario

Bond coupon at t = 1,000 x (TIIE observed at t-0.5 + 2%) / 2

Net IRS at t = 1,000 x (8% - TIIE observed at t-0.5) / 2

Combined cash flow at t = bond cash flow at t + net IRS at t

The cancellation requires the same currency, benchmark, notional and dates. It fixes the scheduled peso income under these assumptions. It does not fix the market value of the position or its dollar return.

When TIIE falls: protect the income

Keep the bond and swap terms unchanged. Only the future TIIE observations change. The table starts at t=0 so that the rate used for the first payment is visible. All cash amounts are MXN; positive numbers are receipts and negative numbers are payments.

Date	TIIE observed at this date	Bond cash flow	Net IRS	Bond + IRS
t=0	8%	-1,000	0	-1,000
t=0.5	6%	50	0	50
t=1.0	4%	40	+10	50
t=1.5	2%	30	+20	50
t=2.0	2%	20	+30	50
t=2.5	2%	20	+30	50
t=3.0	2% (unused)	1,020	+30	1,050

Each cash-flow row uses the rate from the preceding row. The rate observed at t=3.0 is unused because no semester remains. The final bond payment includes principal; the swap has no principal payment.

Follow the payment at t=1.0

Use 6% TIIE observed at t=0.5. The bond coupon is 1,000 x (6% + 2%) / 2 = 40. The swap receives 1,000 x 8% / 2 = 40 and pays 1,000 x 6% / 2 = 30.

Net IRS = 40 - 30 = +10; combined income = 40 + (+10) = 50

Compare the three-year totals

Interest only; excludes principal	Without IRS	With IRS
Total bond coupons	180	180
Total net IRS	0	+120
Total interest income	180	300
Cumulative cash return	18%	30%

These are cumulative cash returns over three years, not annualized yields. Coupons are held without reinvestment. The principal is returned separately and is not counted as interest income.

What I want you to notice

The bond coupons shrink, but the swap receipts increase. The IRS adds MXN 120 over the three years, preserving MXN 50 of interest every semester. The manager has protected income against falling rates.

When TIIE rises: give up the upside

Keep the bond and swap terms unchanged. Only the future TIIE observations change. The table starts at t=0 so that the rate used for the first payment is visible. All cash amounts are MXN; positive numbers are receipts and negative numbers are payments.

Date	TIIE observed at this date	Bond cash flow	Net IRS	Bond + IRS
t=0	8%	-1,000	0	-1,000
t=0.5	10%	50	0	50
t=1.0	12%	60	-10	50
t=1.5	14%	70	-20	50
t=2.0	14%	80	-30	50
t=2.5	14%	80	-30	50
t=3.0	14% (unused)	1,080	-30	1,050

Each cash-flow row uses the rate from the preceding row. The rate observed at t=3.0 is unused because no semester remains. The final bond payment includes principal; the swap has no principal payment.

Follow the payment at t=1.0

Use 10% TIIE observed at t=0.5. The bond coupon is 1,000 x (10% + 2%) / 2 = 60. The swap receives 1,000 x 8% / 2 = 40 and pays 1,000 x 10% / 2 = 50.

Net IRS = 40 - 50 = -10; combined income = 60 + (-10) = 50

Compare the three-year totals

Interest only; excludes principal	Without IRS	With IRS
Total bond coupons	420	420
Total net IRS	0	-120
Total interest income	420	300
Cumulative cash return	42%	30%

These are cumulative cash returns over three years, not annualized yields. Coupons are held without reinvestment. The principal is returned separately and is not counted as interest income.

What I want you to notice

The bond earns more, but the swap requires net payments. The manager gives up MXN 120 of additional income and still receives MXN 50 per semester. The hedge worked as designed: predictable income was the objective, not the highest income in every scenario.

Fixed peso income still leaves dollar risk

Now evaluate the same investment as a U.S. dollar fund. A positive peso return does not guarantee a positive dollar return. The IRS has fixed the interest cash flows in MXN, but the exchange rate determines how many dollars those pesos will buy.

Be explicit about when the currency is converted

For this example, invest MXN 1,000 at t=0, hold all six MXN 50 coupons in pesos without reinvestment, and convert the accumulated MXN 1,300 at t=3. There are no intermediate dollar conversions. The local cumulative return is 30%.

USDMXN means MXN per USD. A rise in USDMXN means peso depreciation: more pesos are required to buy one dollar. To convert a peso amount to dollars, divide it by USDMXN.

	At t=0	At t=3
USDMXN	20	28
MXN investment / proceeds	1,000	1,300
USD investment / proceeds	1,000 / 20 = 50	1,300 / 28 = 46.43

USD return = (1,300 / 28) / (1,000 / 20) - 1 = -7.14%

The fund earned 30% in pesos, but the depreciation was large enough to produce a negative dollar return. Local appreciation would help the dollar result; depreciation reduces it and can reverse its sign.

Find the break-even exchange rate

USD return = 1.30 x $USDMXN_0$ / $USDMXN_3$ - 1

With USDMXN initially at 20, the break-even terminal rate is 1.30 x 20 = 26. At 26, the dollar return is zero. Above 26, it is negative. This threshold applies to the stated assumption that all pesos are converted at the end. If coupons are converted as received, use the exchange rate for each payment date.

Separate the two risks

The receive-fixed IRS removes the matched TIIE exposure from the scheduled peso income. A separate currency hedge would be needed to fix the dollar value of those cash flows. Once the peso amounts are fixed, the manager can identify the amounts and dates to hedge; the IRS alone does not provide that FX protection.

The comparison to remember

Without the IRS, the local interest income is MXN 180 when rates fall and MXN 420 when rates rise. With the IRS, it is MXN 300 in both cases. That is the interest-rate decision. Whether those pesos deliver a positive dollar return is a second question, governed by the FX conversion.